

GOOGLE CLOUD PROFESSIONAL CLOUD ARCHITECT

Section 5.1 — Managing Identity and Access Management (IAM)

GCP Associate Cloud Engineer

Study Notes Study Notes April 2026

Exam Relevance

This topic is part of **Section 5: Configuring access and security (~17.5 % of the exam)**. You must know how to view and create IAM policies, manage the different role types (basic, predefined, custom), and define custom IAM roles.

1. IAM Fundamentals

The IAM Model

WHO (Principal/Member) + WHAT (Role/Permissions) + WHERE (Resource) = IAM Policy

IAM answers three questions: 1. **Who** is making the request? (Member/Principal) 2. **What** are they allowed to do? (Role → Permissions) 3. **On which** resource? (Project, folder, organization, or specific resource)

Principal Types

A **principal** is the identity that is granted access. GCP supports 6 principal types:

Policy prefix	What it represents	Example
<code>user:</code>	An individual Google Account	<code>user:alice@example.com</code>
<code>serviceAccount:</code>	A GCP service account	<code>serviceAccount:my-sa@project.iam.gserviceaccount.com</code>
<code>group:</code>	A Google Group (contains multiple users)	<code>group:devs@example.com</code>
<code>domain:</code>	All users in a Google Workspace / Cloud Identity domain	<code>domain:example.com</code>
<code>allAuthenticatedUsers</code>	Any authenticated Google account in the world	<i>(no prefix)</i>
<code>allUsers</code>	Anyone — including unauthenticated, anonymous access	<i>(no prefix)</i>

Exam tips: - `allUsers` = fully public access (used for public GCS buckets, unauthenticated Cloud Run services) - `allAuthenticatedUsers` ≠ “all users in my organization” — it means any Google account worldwide - `domain:` is the most efficient way to grant access to all employees in a company - Service accounts appear as principals when they are granted roles on resources (e.g., a SA that reads from GCS)

How IAM Policies Work

```

Organization Policy
├── Folder Policy
│   ├── Project Policy
│       └── Resource Policy

```

Effective permissions = Union of all inherited + directly granted permissions

- Policies are **additive** — permissions are unioned, never subtracted
- **Deny policies** (introduced later) can explicitly deny permissions
- Child resources **inherit** parent policies
- A more permissive parent policy **cannot be overridden** by a more restrictive child policy

2. Viewing IAM Policies

Viewing Project-Level IAM Policies

```
# View the IAM policy for a project
gcloud projects get-iam-policy PROJECT_ID

# View in JSON format (for editing)
gcloud projects get-iam-policy PROJECT_ID --format=json

# View in YAML format
gcloud projects get-iam-policy PROJECT_ID --format=yaml

# Filter to see specific member's roles
gcloud projects get-iam-policy PROJECT_ID \
  --flatten="bindings[].members" \
  --filter="bindings.members:alice@example.com" \
  --format="table(bindings.role) "
```

Viewing IAM Policies at Other Levels

```
# Organization level
gcloud organizations get-iam-policy ORG_ID

# Folder level
gcloud resource-manager folders get-iam-policy FOLDER_ID

# Resource level (examples)
gcloud storage buckets get-iam-policy gs://my-bucket
gcloud compute instances get-iam-policy my-vm --zone=us-central1-a
gcloud pubsub topics get-iam-policy my-topic
gcloud run services get-iam-policy my-service --region=us-central1
```

Testing IAM Permissions

```
# Test if a member has specific permissions
gcloud projects test-iam-permissions PROJECT_ID \
  --permissions=compute.instances.create,compute.instances.delete
```

IAM Policy Analyzer

```
# Analyze who has access to a resource
gcloud asset analyze-iam-policy \
  --organization=ORG_ID \
  --identity="user:alice@example.com"

# Find all members with a specific role
gcloud asset analyze-iam-policy \
  --organization=ORG_ID \
  --roles="roles/owner"
```

3. Creating and Modifying IAM Policies

Adding Role Bindings

```
# Grant a role at the project level
gcloud projects add-iam-policy-binding PROJECT_ID \
  --member="user:alice@example.com" \
  --role="roles/compute.admin"

# Grant a role to a group
gcloud projects add-iam-policy-binding PROJECT_ID \
  --member="group:developers@example.com" \
  --role="roles/editor"

# Grant a role to a service account
gcloud projects add-iam-policy-binding PROJECT_ID \
  --member="serviceAccount:my-sa@PROJECT_ID.iam.gserviceaccount.com" \
  --role="roles/storage.objectViewer"

# Grant a role at the organization level
gcloud organizations add-iam-policy-binding ORG_ID \
  --member="user:admin@example.com" \
  --role="roles/resourcemanager.organizationAdmin"

# Grant a role at the folder level
gcloud resource-manager folders add-iam-policy-binding FOLDER_ID \
  --member="group:team@example.com" \
  --role="roles/viewer"
```

Removing Role Bindings

```
# Remove a role binding
gcloud projects remove-iam-policy-binding PROJECT_ID \
  --member="user:alice@example.com" \
  --role="roles/compute.admin"
```

Conditional IAM Bindings

```
# Grant a role with a time-based condition
gcloud projects add-iam-policy-binding PROJECT_ID \
  --member="user:contractor@example.com" \
  --role="roles/compute.instanceAdmin.v1" \
  --condition='expression=request.time < timestamp("2025-12-31T00:00:00Z"),title=temporary'

# Grant access only to resources with specific names
gcloud projects add-iam-policy-binding PROJECT_ID \
  --member="user:dev@example.com" \
  --role="roles/compute.instanceAdmin.v1" \
  --condition='expression=resource.name.startsWith("projects/PROJECT_ID/zones/us-central1-")'

# Grant access based on resource tags
gcloud projects add-iam-policy-binding PROJECT_ID \
  --member="group:team@example.com" \
  --role="roles/compute.admin" \
  --condition='expression=resource.matchTag("env","dev"),title=dev-only'
```

IAM Deny Policies

Deny policies explicitly **prevent** principals from using specific permissions, even if they are granted through allow policies:

```
# Create a deny policy
gcloud iam policies create my-deny-policy \
  --attachment-point="cloudresourcemanager.googleapis.com/projects/PROJECT_ID" \
  --kind=denypolicies \
  --policy-file=deny-policy.json
```

```
{
  "displayName": "Deny project deletion",
  "rules": [
    {
      "denyRule": {
        "deniedPrincipals": ["principalSet://goog/group/developers@example.com"],
        "deniedPermissions": ["cloudresourcemanager.googleapis.com/projects.delete"]
      }
    }
  ]
}
```

4. Role Types

Basic (Primitive) Roles

Role	Name	Description
<code>roles/viewer</code>	Viewer	Read-only access to all resources
<code>roles/editor</code>	Editor	Read-write access (no IAM management)
<code>roles/owner</code>	Owner	Full access including IAM and billing
<code>roles/browser</code>	Browser	Browse project hierarchy

Important for the exam: - Basic roles are **very broad** — they grant thousands of permissions - **Google recommends predefined roles** instead of basic roles - **Owner** includes Editor + IAM admin + billing management - **Editor** does NOT include permission to modify IAM policies - **Never use basic roles in production** unless absolutely necessary

Predefined Roles

Google-managed roles with fine-grained, service-specific permissions:

Category	Example Roles
Compute Engine	roles/compute.admin , roles/compute.instanceAdmin.v1 , roles/compute.networkAdmin , roles/compute.viewer
Cloud Storage	roles/storage.admin , roles/storage.objectViewer , roles/storage.objectCreator
BigQuery	roles/bigquery.admin , roles/bigquery.dataEditor , roles/bigquery.jobUser , roles/bigquery.user
GKE	roles/container.admin , roles/container.developer , roles/container.viewer
Cloud Run	roles/run.admin , roles/run.developer , roles/run.invoker
Cloud SQL	roles/cloudsql.admin , roles/cloudsql.editor , roles/cloudsql.viewer
IAM	roles/iam.serviceAccountAdmin , roles/iam.serviceAccountUser , roles/iam.workloadIdentityUser
Logging	roles/logging.admin , roles/logging.viewer , roles/logging.logWriter
Monitoring	roles/monitoring.admin , roles/monitoring.viewer , roles/monitoring.editor

```
# List all predefined roles
gcloud iam roles list

# List roles for a specific service
gcloud iam roles list --filter="name:roles/compute"

# Describe a role (see all permissions)
gcloud iam roles describe roles/compute.instanceAdmin.v1

# List permissions in a role
gcloud iam roles describe roles/storage.objectViewer \
  --format="value(includedPermissions) "
```

Custom Roles

When predefined roles don't match your needs, create **custom roles** with exactly the permissions required (least privilege):

```
# Create a custom role at the project level
gcloud iam roles create myCustomRole \
  --project=PROJECT_ID \
  --title="My Custom Role" \
  --description="Custom role for app deployment" \
  --permissions=compute.instances.get,compute.instances.list,storage.objects.get,storage.c
  --stage=GA

# Create a custom role at the organization level
gcloud iam roles create orgCustomRole \
  --organization=ORG_ID \
  --title="Organization Custom Role" \
  --permissions=resourcemanager.projects.list,resourcemanager.folders.list

# Create from a YAML file
gcloud iam roles create myRole \
  --project=PROJECT_ID \
  --file=role-definition.yaml
```

role-definition.yaml:

```
title: "App Deployer"
description: "Can deploy and manage applications"
stage: "GA"
includedPermissions:
  - compute.instances.get
  - compute.instances.list
  - compute.instances.create
  - compute.instances.delete
  - storage.objects.get
  - storage.objects.list
  - storage.objects.create
```


Managing Custom Roles

```
# List custom roles
gcloud iam roles list --project=PROJECT_ID

# Describe a custom role
gcloud iam roles describe myCustomRole --project=PROJECT_ID

# Update a custom role (add permissions)
gcloud iam roles update myCustomRole \
  --project=PROJECT_ID \
  --add-permissions=compute.instances.start,compute.instances.stop

# Update a custom role (remove permissions)
gcloud iam roles update myCustomRole \
  --project=PROJECT_ID \
  --remove-permissions=compute.instances.delete

# Disable a custom role
gcloud iam roles update myCustomRole \
  --project=PROJECT_ID \
  --stage=DISABLED

# Delete a custom role (soft delete — can be undeleted within 7 days)
gcloud iam roles delete myCustomRole --project=PROJECT_ID

# Undelete a custom role
gcloud iam roles undelete myCustomRole --project=PROJECT_ID
```

Custom Role Stages

Stage	Description
ALPHA	Role is in early testing
BETA	Role has been tested, may change
GA	Role is generally available and stable
DISABLED	Role is disabled (cannot be granted)

Custom Role Limitations

- Maximum **3,000 permissions** per custom role
- Maximum custom roles per organization or project (varies)
- Some permissions **cannot be used in custom roles** (check documentation)
- Custom roles can only be created at the **project or organization level** (not folder)

5. IAM Best Practices

Principle of Least Privilege

- Grant the **minimum permissions** needed for the task
- Use **predefined roles** instead of basic roles
- Use **custom roles** when predefined roles are too broad
- Review and audit permissions regularly

Use Groups, Not Individual Users

- Create Google Groups for teams/functions
- Grant roles to groups, not individual users
- Easier to manage, audit, and revoke access

Separation of Duties

- Don't give one person Owner on everything
- Separate network admin, security admin, and billing admin
- Use different service accounts for different applications

Audit and Review

- Regularly review IAM policies
- Use **IAM Recommender** to identify over-provisioned access
- Enable **Data Access audit logs** for sensitive resources
- Use **Policy Analyzer** to understand who has access to what

```
# Get IAM recommendations
gcloud recommender recommendations list \
  --project=PROJECT_ID \
  --recommender=google.iam.policy.Recommender \
  --location=global
```

6. IAM Policy Structure

Policy JSON Format

```
{
  "bindings": [
    {
      "role": "roles/storage.objectViewer",
      "members": [
        "user:alice@example.com",
        "group:readers@example.com",
        "serviceAccount:my-sa@project.iam.gserviceaccount.com"
      ]
    },
    {
      "role": "roles/storage.objectAdmin",
      "members": [
        "user:admin@example.com"
      ],
      "condition": {
        "title": "production-only",
        "description": "Only for production bucket",
        "expression": "resource.name.startsWith('projects/_/buckets/prod-')"
      }
    }
  ],
  "etag": "BwXXXXXXXXXX=",
  "version": 3
}
```

Key Fields

- **bindings** — List of role-to-members mappings
 - **role** — The IAM role being granted
 - **members** — List of identities receiving the role
 - **condition** — Optional CEL expression for conditional access
 - **etag** — Concurrency control (prevents race conditions when updating)
 - **version** — Policy version (3 for conditional bindings)
-

7. User Suspension

What “Suspended” Means

A **suspended user** is a Google Workspace (formerly G Suite) account that has been temporarily disabled by a Workspace admin. Suspension is not a GCP-native IAM concept — it lives at the Google identity layer, one level below GCP IAM.

When a user is suspended:

- They **cannot sign in** to any Google service (Gmail, Console, gcloud CLI, etc.)
- Their **IAM bindings in GCP are not removed** — they still appear in every `get-iam-policy` output
- Their data, Drive files, and settings are **preserved** unchanged
- They can be **reinstated** by a Workspace admin at any time

Why It Matters for IAM

Even though a suspended user cannot authenticate, their IAM role bindings remain active in GCP. This has practical consequences:

- If reinstated, they immediately regain all their previous GCP permissions — no re-granting needed
- IAM Recommender may flag their bindings as unused, prompting a review
- Policy Analyzer will still list them as having access to resources

Best practice: when a user is suspended (e.g., offboarding), **also remove their IAM bindings** manually. Don't rely on suspension alone as an access control measure.

Suspension vs. Deletion

State	Can sign in	IAM bindings	Data	Reversible
Active	Yes	Active	Present	—
Suspended	No	Remain (inactive)	Preserved	Yes — admin reinstates
Deleted	No	Cleaned up by Google eventually	Deleted (20-day recovery window)	Within 20 days only

Service Account Disabling (Analogous Concept)

GCP’s equivalent for service accounts is **disabling** (not deleting):

```
# Disable a service account (blocks all authentication, bindings remain)
gcloud iam service-accounts disable SA_EMAIL

# Re-enable a service account
gcloud iam service-accounts enable SA_EMAIL

# Check current status
gcloud iam service-accounts describe SA_EMAIL --format="value(disabled) "
```

A disabled service account behaves exactly like a suspended user: authentication fails, but all IAM role bindings on resources remain intact and are instantly restored when re-enabled.

Exam Angle

The exam may describe a scenario where an employee goes on leave or is offboarded and ask the safest/most reversible action. The answer hierarchy: 1. **Suspend** the Workspace account → blocks login, preserves everything, fully reversible 2. **Remove IAM bindings** → ensures no access even if reinstated unexpectedly 3. **Delete** the account → only when certain they won't return (20-day window to undo)

For service accounts, prefer **disabling** over deleting when the SA may be needed again.

8. Organization Policies (Constraints)

- Organization policies enforce guardrails across an entire organization, folder, or project — regardless of IAM bindings
- Managed by the Organization Policy Administrator (`roles/orgpolicy.policyAdmin`)
- Two types of constraints: **boolean** (on/off) and **list** (allow/deny specific values)

Common constraints

```
# Disable VM external IP (list constraint)
gcloud resource-manager org-policies set-policy \
  --project=PROJECT_ID \
  --policy='{
    "constraint": "constraints/compute.vmExternalIpAccess",
    "listPolicy": {"allValues": "DENY"}
  }'

# Require OS Login (boolean)
gcloud resource-manager org-policies enable-enforce \
  --project=PROJECT_ID \
  constraints/compute.requireOsLogin

# Restrict allowed cloud regions
gcloud resource-manager org-policies set-policy \
  --organization=ORG_ID \
  --policy='{
    "constraint": "constraints/gcp.resourceLocations",
    "listPolicy": {"allowedValues": ["in:us-locations"]}
  }'

# View effective policy (merging parent and current level)
gcloud resource-manager org-policies describe constraints/compute.vmExternalIpAccess --project=PROJECT_ID
gcloud resource-manager org-policies list --organization=ORG_ID
```

Common exam constraints to know

Constraint	Purpose
<code>compute.vmExternalIpAccess</code>	Prevent VMs from getting public IPs
<code>compute.requireOsLogin</code>	Enforce OS Login on all VMs
<code>compute.restrictCloudSQLInstances</code>	Prevent Cloud SQL from having public IP
<code>iam.allowedPolicyMemberDomains</code>	Restrict IAM to specific domains
<code>gcp.resourceLocations</code>	Limit which regions resources can be created in
<code>storage.uniformBucketLevelAccess</code>	Enforce uniform bucket-level access on GCS
<code>iam.disableServiceAccountKeyCreation</code>	Prevent SA key downloads

Inheritance: org-level policies are inherited by all children; a child can only restrict further (cannot override to be more permissive unless the org policy itself allows override).

9. IAM Recommender

- Automatically analyzes permissions usage over **90 days** and suggests removing unused permissions
- Recommendations are per-member-per-role-per-resource
- Each recommendation has an impact score (HIGH, MEDIUM, LOW)

```
# List IAM recommendations for a project
gcloud recommender recommendations list \
  --project=PROJECT_ID \
  --recommender=google.iam.policy.Recommender \
  --location=global \
  --format="table(name,stateInfo.state,priority,description)"

# Apply a recommendation (replace role with smaller one)
gcloud recommender recommendations mark-claimed RECOMMENDATION_ID \
  --project=PROJECT_ID \
  --recommender=google.iam.policy.Recommender \
  --location=global \
  --etag=ETAG

# Mark as dismissed (won't apply)
gcloud recommender recommendations mark-dismissed RECOMMENDATION_ID \
  --project=PROJECT_ID --recommender=google.iam.policy.Recommender --location=global
```

Exam tip: Recommender suggests replacing `roles/editor` with specific predefined roles. Always review before applying — some roles may be used programmatically without being captured in 90-day logs.

Exam Practice Questions

1. A developer needs to create and manage Compute Engine VMs but should not be able to modify firewall rules or networking. Which role should you grant?
 - Answer: `roles/compute.instanceAdmin.v1` — Grants full control over VMs but not networking resources.

2. **You need to grant a contractor temporary access to view Cloud Storage objects, expiring December 31. How?**

- Answer: Use a **conditional IAM binding** with a time expression: `--condition='expression=request.time < timestamp("2025-12-31T00:00:00Z"),title=temp-access'` with `roles/storage.objectViewer`.

3. **A team of 15 developers needs identical access to GKE. What's the most efficient way to manage this?**

- Answer: Create a **Google Group** (e.g., `gke-devs@example.com`), add all 15 developers, and grant `roles/container.developer` to the group at the project level.

4. **The predefined `roles/storage.admin` role grants too many permissions for your use case. You only need object read and bucket list permissions. What should you do?**

- Answer: Create a **custom role** with only the specific permissions needed: `storage.objects.get`, `storage.objects.list`, `storage.buckets.list`.

5. **A user has `roles/viewer` at the organization level and `roles/compute.admin` at a specific project. What are their effective Compute Engine permissions in that project?**

- Answer: They have **both** — `roles/viewer` (inherited from org) AND `roles/compute.admin` (directly on project). IAM policies are additive, so they get the union of all permissions.

6. **Can you remove `roles/viewer` permissions from a user at the project level if they were granted `roles/editor` at the folder level?**

- Answer: **No**. IAM policies are additive and inherited. You cannot override a parent policy at a child level using allow policies. You would need to use a **deny policy** or remove the binding at the folder level.

7. **What is the difference between `roles/editor` and `roles/owner` ?**

- Answer: `roles/editor` grants read-write access to most resources but **cannot manage IAM policies or billing**. `roles/owner` grants all Editor permissions **plus** IAM management and billing management.

Glossary

allAuthenticatedUsers: A special IAM principal identifier that represents any user who is authenticated with a Google account worldwide; it does not restrict access to users within a specific organization.

allUsers: A special IAM principal identifier that grants access to anyone, including unauthenticated (anonymous) users; used to make resources such as Cloud Storage buckets or

Cloud Run services fully public.

Basic role: A coarse-grained, legacy IAM role (`roles/viewer` , `roles/editor` , `roles/owner` , `roles/browser`) that grants broad permissions across all GCP services; Google recommends using predefined or custom roles instead.

Boolean constraint: An organization policy constraint type that is either enforced (on) or not enforced (off), such as `compute.requireOsLogin` .

CEL (Common Expression Language): A lightweight expression language used in conditional IAM bindings to evaluate attributes such as request time, resource name, and resource tags at policy evaluation time.

Cloud Identity: Google's Identity-as-a-Service platform that provides managed user and group identities for organizations that do not use Google Workspace; supports `domain:` principal type in IAM.

Cloud SQL: Google Cloud's fully managed relational database service; referenced in organization policy constraints such as `compute.restrictCloudSQLInstances` to prevent public IP assignment.

Compute Engine: Google Cloud's Infrastructure-as-a-Service (IaaS) offering for running virtual machines; a primary resource type used in IAM policy examples and organization policy constraints.

Condition (IAM): An optional CEL expression attached to an IAM role binding that restricts when and where the binding is effective, based on attributes such as request time, resource name, or resource tags.

Custom role: A user-defined IAM role created at the project or organization level that contains exactly the permissions required for a specific use case, supporting the principle of least privilege.

Deny policy: An IAM policy type that explicitly prevents specified principals from using specified permissions, overriding any allow policies that would otherwise grant those permissions.

domain: An IAM principal type prefix (`domain:example.com`) that grants access to all users within a specific Google Workspace or Cloud Identity domain; the most efficient way to grant access to all employees in a company.

Editor: A basic IAM role (`roles/editor`) that grants read-write access to most GCP resources but does not include IAM management or billing administration permissions.

etag: A concurrency control token included in IAM policy JSON that prevents race conditions when multiple clients read and update the same policy simultaneously; must be included when writing back a policy.

Folder: A GCP resource hierarchy node below the organization and above projects, used to group related projects and apply inherited IAM policies and organization policies to multiple projects at once.

gcloud asset analyze-iam-policy: A CLI command that uses Policy Analyzer to determine which principals have access to which resources, useful for security audits and access reviews.

gcloud recommender: A CLI command group used to list and act on IAM Recommender suggestions for removing or reducing over-provisioned IAM role bindings.

Google Group: A collection of Google account identities that can be used as a single IAM principal (`group:`); granting roles to groups rather than individual users simplifies access management.

Google Workspace: Google's cloud-based productivity and collaboration suite (formerly G Suite); provides user identities that can be suspended, deleted, or reinstated by a Workspace admin.

group: An IAM principal type prefix (`group:`) that grants access to all members of a specified Google Group, enabling centralized management of permissions for a team or function.

IAM (Identity and Access Management): Google Cloud's system for controlling which principals can perform which actions on which resources, enforced through policies consisting of role bindings at various resource hierarchy levels.

IAM policy: A document attached to a GCP resource (organization, folder, project, or individual resource) that specifies a set of role bindings, each mapping a role to one or more principals.

IAM policy binding: A single entry in an IAM policy that associates a role with one or more principals, optionally constrained by a condition.

IAM Recommender: A Google Cloud service that analyzes IAM permission usage over a 90-day window and automatically generates recommendations to remove or reduce unused or over-provisioned role grants.

IAM role: A named collection of permissions that can be granted to a principal; roles are either basic (primitive), predefined (Google-managed), or custom (user-defined).

Inheritance (IAM): The mechanism by which IAM policies applied at a parent resource level (organization or folder) are automatically granted to all descendant resources (folders, projects, and resources within them).

List constraint: An organization policy constraint type that enforces an allow or deny list of specific values, such as permitted resource locations or allowed project member domains.

Organization: The root node of the GCP resource hierarchy representing a company or entity, to which organization-wide IAM policies and organization policies are applied.

Organization policy: A governance control applied at the organization, folder, or project level that enforces guardrails (constraints) on resource configuration independently of IAM role bindings.

Organization Policy Administrator: The IAM role (`roles/orgpolicy.policyAdmin`) that grants the ability to view and set organization policies across the resource hierarchy.

Owner: A basic IAM role (`roles/owner`) that grants all Editor permissions plus the ability to manage IAM policies and billing; the most permissive basic role.

Policy Analyzer: A GCP tool (accessible via `gcloud asset analyze-iam-policy`) that answers access questions such as which principals have a specific permission on a resource, used for auditing and compliance.

Predefined role: A Google-managed IAM role with fine-grained, service-specific permissions (e.g., `roles/compute.instanceAdmin.v1`) that is updated automatically when new features are released; preferred over basic roles.

Principal: An identity in Google Cloud IAM that can be granted roles; principal types include user, serviceAccount, group, domain, allAuthenticatedUsers, and allUsers.

Principle of least privilege: The security practice of granting only the minimum permissions required for a task, reducing the blast radius of compromised credentials or misconfigured access.

Resource hierarchy: The organizational structure of GCP resources consisting of organization → folders → projects → resources; IAM policies are inherited down this hierarchy.

Role: A named collection of IAM permissions that defines what actions a principal can perform on GCP resources; roles are the unit of access grant in IAM and come in three types: basic, predefined, and custom.

Role binding: The association of a role to one or more principals within an IAM policy, optionally scoped by a condition; role bindings are additive — permissions are unioned across all bindings that apply to a principal.

roles/browser: A basic IAM role that grants permission to browse the GCP project hierarchy without granting access to any resource data.

roles/compute.admin: A predefined IAM role that grants full control over all Compute Engine resources including VMs, networks, and firewall rules.

roles/compute.instanceAdmin.v1: A predefined IAM role that grants full control over Compute Engine VM instances but not over networking resources such as firewall rules.

roles/compute.networkAdmin: A predefined IAM role that grants full control over Compute Engine networking resources (VPCs, subnets, firewall rules, routes) but not over VM instances.

roles/compute.viewer: A predefined IAM role that grants read-only access to all Compute Engine resources.

roles/container.admin: A predefined IAM role that grants full control over GKE clusters and their workloads.

roles/container.developer: A predefined IAM role that grants access to deploy and manage workloads within existing GKE clusters without cluster administration rights.

roles/container.viewer: A predefined IAM role that grants read-only access to GKE clusters and their workloads.

roles/editor: See Editor.

roles/iam.serviceAccountAdmin: A predefined IAM role that grants full management of service accounts within a project, including creation, deletion, and key management.

roles/iam.serviceAccountUser: A predefined IAM role that allows a principal to attach a service account to a resource such as a VM or Cloud Run service.

roles/iam.workloadIdentityUser: A predefined IAM role granted to external identities (via Workload Identity Federation) to allow them to impersonate a Google Cloud service account.

roles/logging.admin: A predefined IAM role that grants full control over Cloud Logging resources including log buckets, sinks, and metrics.

roles/logging.logWriter: A predefined IAM role that allows a principal to write log entries to Cloud Logging; commonly granted to the writer identity of a log sink.

roles/logging.viewer: A predefined IAM role that grants read-only access to log entries in Cloud Logging.

roles/monitoring.admin: A predefined IAM role that grants full control over Cloud Monitoring resources including alert policies, dashboards, and notification channels.

roles/monitoring.editor: A predefined IAM role that grants read-write access to Cloud Monitoring resources excluding IAM policy management.

roles/monitoring.viewer: A predefined IAM role that grants read-only access to Cloud Monitoring dashboards, alert policies, and metrics.

roles/orgpolicy.policyAdmin: A predefined IAM role that grants the ability to view and set organization policies across the GCP resource hierarchy.

roles/owner: See Owner.

roles/resourcemanager.organizationAdmin: A predefined IAM role that grants full control over the GCP organization resource, including managing IAM policies at the organization level.

roles/run.admin: A predefined IAM role that grants full control over Cloud Run services and their IAM policies.

roles/run.developer: A predefined IAM role that grants the ability to deploy and manage Cloud Run services without modifying IAM policies.

roles/run.invoker: A predefined IAM role that allows a principal to invoke (send HTTP requests to) a Cloud Run service.

roles/storage.admin: A predefined IAM role that grants full control over Cloud Storage buckets and objects, including IAM policy management.

roles/storage.hmacKeyAdmin: A predefined IAM role that grants the ability to create and manage HMAC keys for Cloud Storage, used for S3-compatible API access.

roles/storage.objectAdmin: A predefined IAM role that grants full control over Cloud Storage objects (read, write, delete) but not bucket-level configuration.

roles/storage.objectCreator: A predefined IAM role that allows creating Cloud Storage objects without the ability to read or delete them.

roles/storage.objectViewer: A predefined IAM role that grants read-only access to Cloud Storage objects and their metadata.

roles/viewer: A basic IAM role (`roles/viewer`) that grants read-only access to all GCP resources within the scope it is applied.

Separation of duties: A security principle that divides administrative responsibilities among multiple accounts or roles to prevent any single identity from having excessive control.

serviceAccount: An IAM principal type prefix (`serviceAccount:`) used to reference a Google Cloud service account as a member in an IAM policy binding.

Service account (as a principal): A service account acting as a member in an IAM policy, receiving role bindings that define what it can do on GCP resources.

Service account (as a resource): A service account treated as a GCP resource that has its own IAM policy, controlling which users or other service accounts can use or manage it.

Soft delete: The behavior when a custom IAM role or service account is deleted; it enters a deleted state and can be undeleted within 7 days (custom roles) or 30 days (service accounts) before being permanently removed.

Suspended user: A Google Workspace account that has been temporarily disabled by a Workspace administrator; the user cannot authenticate, but their GCP IAM role bindings remain in place until manually removed.

Tag (resource tag): A key-value label attached to GCP resources that can be referenced in conditional IAM bindings (`resource.matchTag`) to restrict access to resources with specific tags.

user: An IAM principal type prefix (`user:`) that references an individual Google account by email address for use in IAM policy bindings.

Version (IAM policy): The schema version of an IAM policy JSON document; version 3 is required when the policy contains conditional role bindings.

Viewer: A basic IAM role (`roles/viewer`) that grants read-only access to all resources within the scope it is granted; does not allow modification of any resources.

VPC Service Controls: A GCP security feature that creates a service perimeter around GCP resources to prevent data exfiltration; violations are recorded in Policy Denied audit logs.

